

Math 140 Spring 2023 - Worksheet 4.6

Section:

Name:

1. Find the intervals on which the graph of the function $f(x) = x\sqrt{1-x^2}$ is concave upward and on which it is concave downward. Sketch the graph of the function by hand.
2. Find all inflection points (if any) of the graph of the function $f(x) = 3x^4 + 4x^3$. Then sketch the graph of f by hand.
3. Let $g(x) = x + \cos x$. Discuss the concavity of g and sketch it.